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Stochastic simulation of chemically activated unimolecular reactions

Luc Vereecken, Guido Huyberechts,^{a)} and Jozef Peeters^{b)}

Department of Chemistry, University of Leuven, Celestijnenlaan 200 F, B-3001 Leuven, Belgium

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The kinetic master equation for the title processes can be formulated as a traditional deterministic set of coupled differential reaction-rate equations, or, alternatively, as a stochastic process in which each reaction is a random-walk transition in energy-species space. This stochastic description is the basis for three methods we describe here to numerically solve the kinetic master equation for chemically activated unimolecular reactions. The first method allows the calculation of the complete time evolution within a given mechanism, and is based on Gillespie's exact stochastic method (ESM). It is essentially a Monte Carlo simulation of the stochastic reaction processes. The second method allows for the direct calculation of the steady-state product distribution (DCPD). It describes the random walk within the framework of a discrete time Markov chain, and reduces the calculation of the steady-state product distribution to a fairly simple matrix algebra problem. The third method calculates the steady-state population of the intermediates (CSSPI), reformulating the solution of the master equation as an eigenvector problem generated by the description as a continuous time Markov chain. To our knowledge, the DCPD method has not been described before. Also, this is the first time that a CSSPI model is used explicitly in this type of calculation. The three methods are illustrated using the simple $\text{H} + \text{HNCO}$ reaction, important in the RAPRENO_x mechanism for NO_x removal from flue gases. © 1997 American Institute of Physics. [S0021-9606(97)50316-3]

I. INTRODUCTION

Chemically activated unimolecular reactions are an important class of reactions, especially in atmospheric and combustion chemistry. Association of two reactants initially forms an energized adduct, which then undergoes isomerization reactions until one of the isomeric forms eventually dissociates into multiple fragments or is stabilized due to collisional energy transfer reactions with bath gas molecules. While the energy-dependent unimolecular rate constants for these isomerization and dissociation reactions can be readily computed, e.g., by the RRKM theory,¹⁻⁴ solving the master equation to calculate pressure and temperature dependence of the product distribution is often troublesome.

Traditionally, a master equation is written as a set of coupled kinetic differential equations describing the continuous change through time of the concentrations of the participating entities X_i , each of these being a chemical species with given vibrational and rotational energy contents:

$$\frac{d[X_i]}{dt} = f([X_1], [X_2], \dots, [X_n]) \quad \text{for all } i.$$

Simultaneously integrating these equations yields the concentrations of each species X_i at a given time. Generally, however, it is not possible to find an analytical solution for this set of differential equations; one resorts to numerical approximations to obtain the concentrations sought. Due to the specific characteristics of chemically activated unimolecular reaction mechanisms, these numerical integrations are often computationally difficult. These difficulties arise

primarily from limited numerical precision, inducing cumulative (round-off) errors through stiffness of the set of equations, and through the approximation of an infinitesimal time interval dt by a discrete time jump Δt .

Rather than describing the time evolution of a chemically reacting system as both continuous and deterministic, one can also model any chemical system as a discrete stochastic process.⁵⁻⁸ While the stochastic master equation is even harder to solve analytically than its deterministic counterpart—since the latter only attempts to describe the *average* of the true physical random process—one can use the stochastic model to numerically simulate the time evolution of the chemical system. Gillespie⁵⁻⁸ has developed the exact stochastic method (ESM), a Monte Carlo simulation technique based on this stochastic formalism. The algorithm for this simulation is relatively simple, and much less susceptible to cumulative errors than numerical solutions for the classic master equation.

Below, we give a brief description of Gillespie's ESM. Next, we discuss its application to chemically activated unimolecular reaction mechanisms, and its implementation in the URESAM computer program. When opting for a stochastic description to model the time evolution of these reactions, rather than using the equivalent classic master equation, one avoids the computational complexity of numerically solving the several hundreds of coupled differential equations describing chemically activated reactions.

In addition to this general method, we also present two faster alternatives, also implemented in the URESAM program, for calculations under steady-state conditions, both derived on stochastic arguments. The first of these methods (DCPD) directly calculates the product distribution for a given (set of) initial reaction(s), the second method (CSSPI) obtains the (quasi)steady-state concentrations of the acti-

^{a)}Current address: Interuniversity Micro-Electronics Center IMEC, Leuven, Belgium.

^{b)}Author to whom correspondence should be addressed.

vated intermediates, from which the product distribution is then derived. Finally, we illustrate and compare the three methods by means of an example. We assume basic knowledge of stochastic processes and Markov chains.⁹ All three methods are independent of the rate theory used to calculate the unimolecular rate constants and energy transfer probabilities.

II. THE EXACT STOCHASTIC METHOD (ESM)

For a general discussion of the ESM, we refer to the original article and subsequent publications by Gillespie.⁵⁻⁸ The ESM has been used before¹⁰ for calculations on unimolecular reactions, though in a different implementation than ours. Here, we give a detailed description of the ESM and of its implementation for unimolecular reactions; we also address the conditions for the applicability of the ESM using only time-independent rate constants.

All calculations are performed on a discrete number of molecules; conversion of concentration-dependent parameters to number-dependent parameters is straightforward for a homogeneous mixture with a known volume V . Using a unit volume simplifies the implementation.

A. Simulation of the stochastic time evolution

The time evolution of any given chemical system can be simulated if, at any given time t , one can calculate *when* the next reaction will occur, and *which* reaction that will be. By the very nature of chemical reactions, the answer to these two questions is preferentially described as a stochastic process. The probability that the next reaction will occur in the time interval $[t + \tau, t + \tau + d\tau]$, and that it will be reaction μ , can be calculated as the product of the probability that no reaction has occurred yet in the time interval $[t, t + \tau]$, and the probability that the subsequent reaction, occurring in the time interval $[t + \tau, t + \tau + d\tau]$, will be reaction μ . Both the time lapse τ until the next reaction, as well as the exact nature μ of the reaction occurring are therefore random quantities drawn from their respective probability distributions. Given the two corresponding probability density functions $P(\tau)$ and $P(\mu)$, the stochastic time evolution of a given chemical system can be simulated by repetitively generating random time intervals τ and reactions μ , according to these probability density functions $P(\tau)$ and $P(\mu)$. Each time, the simulated time t is incremented to $t + \tau$, and the populations are adapted to reflect the occurrence of reaction μ . The functions $P(\tau)$ and $P(\mu)$ can be derived from the population at the given time t , and the reaction rate constants by calculating, for every reaction μ , the number h_μ of distinct reactant combinations per volume at time t , and the stochastic reaction probability constant c_μ . This reaction probability constant c_μ is equal to the deterministic rate constant k_μ , converted to a number-dependent parameter:

$$c_\mu = k_\mu V^{n-1} \prod_{i=1}^{n_\mu} M_i!$$

with n the molecularity of reaction μ , n_μ the number of different reactant species in reaction μ , and M_i the partial

molecularity for each reactant i . h_μ is the discrete counterpart of the concentration factors in classical kinetic expressions, and is readily derived by combinatorial arguments. In general, the number of distinct reactant combinations is given by

$$h_\mu = \prod_{i=1}^{n_\mu} \frac{N_i!}{M_i!(N_i - M_i)!},$$

where N_i is the number of molecules of reactant i present in volume V . For reactions with low molecularity, this expression becomes very simple; e.g., for a unimolecular reaction $S \rightarrow T$, it yields $h_\mu = N_S$.

B. Derivation of $P(\tau)$ and $P(\mu)$

If we can model the chemical reaction μ by a Poisson process with intensity $h_\mu c_\mu$ —the basic assumption of this simulation algorithm—derivation of expressions for $P(\tau)$ and $P(\mu)$ is fairly straightforward. The process describing all n reactions simultaneously is the sum of these Poisson processes, and is a Poisson process in itself, with intensity $\sum_{i=1}^n h_i c_i$. It is readily shown that the probability that no process will occur during the following period τ is equal to $\exp(-\sum_{i=1}^n h_i c_i \tau)$. Therefore, the probability at time t that the next occurrence will take place during the time interval $t + \tau$ to $t + \tau + d\tau$ is given by

$$P(\tau) d\tau = \sum_{i=1}^n h_i(t) c_i \times \exp\left(-\sum_{i=1}^n h_i(t) c_i \tau\right) d\tau.$$

The probability that this first occurrence after a time lapse τ will be Poisson process μ is given by

$$P(\mu) = \frac{h_\mu(t) c_\mu}{\sum_{i=1}^n h_i(t) c_i}.$$

Since both $P(\tau)$ and $P(\mu)$ depend on the current population—through $h_i(t)$ —they have to be recalculated after each reaction. Note also that $P(\mu)$ is not a function of τ due to the memory-free property of the exponential distribution. Generating random time intervals τ and reactions μ from these probability density functions can be accomplished with the inversion method. First, we construct the probability distribution functions $F_\tau(\tau_r)$ and $F_\mu(\mu_r)$, which give the cumulative probability for obtaining a value of τ smaller than or equal to τ_r , and a value of μ smaller than or equal to μ_r , respectively; F_τ and F_μ always take their values in the interval $[0..1]$:

$$F_\tau(\tau_r) = 1 - \exp\left(-\sum_{i=1}^n h_i c_i \tau_r\right);$$

$$F_\mu(\mu_r) = \left(\frac{1}{\sum_{i=1}^n h_i c_i}\right) \times \sum_{i=1}^{\mu_r} h_i c_i.$$

Using a random number generator subroutine, we can generate random numbers r_1 and r_2 , uniformly distributed in the interval $[0..1]$. The random time lapse τ_r can then be calculated by inverting $F_\tau(\tau_r) = r_1$, taking into account that r_1 and $(1 - r_1)$ are both uniformly distributed:

$$\tau_r = \frac{1}{\sum_{i=1}^n h_i c_i} \ln \left(\frac{1}{r_1} \right).$$

Likewise, the reaction occurring after that time period is given by the integer number μ_r for which:

$$\sum_{i=1}^{\mu_r-1} h_i c_i < r_2 \times \sum_{i=1}^n h_i c_i \leq \sum_{i=1}^{\mu_r} h_i c_i.$$

The resulting random variables τ_r and μ_r will be distributed according to $P(\tau)$ and $P(\mu)$, respectively.

C. Discussion

In Gillespie's article,⁶ the fundamental hypothesis was that the average probability of a reaction is equal to $c_\mu dt$. Gillespie states that the condition for which this requirement is fulfilled is that "nonreactive molecular collisions occur much more frequently than reactive molecular collisions." Next to being defined somewhat vaguely, this condition is too strict. Having more nonreactive than reactive collisions implies that all species obey a Maxwell-Boltzmann energy distribution, ensuring that a reaction works on a constant population distribution and therefore always has the same average probability (proportional to) $c_\mu dt$. This kind of process can obviously be modeled by a Poisson process with intensity $h_\mu c_\mu$. However, a Maxwell-Boltzmann distribution is not necessary for this algorithm to work, only to allow a population-averaged description of the chemical reactions. As long as all considered reactions can be modeled by Poisson processes, or can be split up into Poisson processes, the algorithm holds. An example of this is the simulation of chemically activated unimolecular reaction systems, the subject of this article. Since the adducts are formed initially with a high energy content, which they gradually lose in collisions with the bath gas, the energy distribution functions—and therefore the average unimolecular rate constants—are functions of time. However, if we narrow down the definition of one particular chemical entity S to include its specific energy content E , we can calculate time independent reaction probabilities $c_\mu(E)$ working on one particular $S(E)$. If we also include all possible energy transfer reactions, we can accurately simulate the unimolecular reaction systems, even at very low pressures. This can be further extended to account for different rotational quantum numbers J , etc. The same approach can be applied in principle also to nonunimolecular reactions.

Being able to describe all reactions as Poisson processes—reactions with a time-independent rate constant—is a desirable property for this type of simulation. It can be proven mathematically⁹ that a time-independent rate constant (math: "hazard-rate") always leads to an exponential probability distribution function, and vice versa. Exponential distributions can be readily combined to describe multiple (parallel) possible reactions, always leading to *time-independent relative probabilities* for the individual components (see Sec. II B), and a simple expression for the inter-reaction time lapses. Including time-dependent rate constants results in more complicated expressions for which the rela-

tive probabilities are time dependent, excepting degenerate cases for which *all* rate constants scale identically in time. Selection of the inter-reaction time lapses also becomes more complex; often a closed expression cannot be found, so numerical integration or root solving must be used to obtain properly distributed random time lapses. Obviously, this has an adverse impact on the calculation time needed for a simulation, and on the ease of implementation. Additionally, if the change of the rate constant depends on the running lifetime of the molecule (e.g., changing probability of unimolecular dissociation because of slow energy redistribution), rather than being caused by the change of a global factor (e.g., the changing light intensity for photochemical reactions), it becomes necessary to keep track of the moment of formation of each individual molecule. The memory-free property of the exponential distribution—resulting from the use of time-independent rate constants only—allows one to ignore this information and perform calculations based only on the current population.

D. Implementation of the exact stochastic method in URESAM

Based on this exact stochastic method (ESM), the URESAM computer program was developed to calculate the product distribution of chemically activated unimolecular reaction systems. As discussed above, the energy scale for each different chemical species is divided into quantized energy bands, small enough to consider all properties—including rate constants—within one energy band as constant. This allows for time-dependent energy distributions; in the present version of the program, only different vibrational energies are considered. As far as the algorithm is concerned, the molecules present in each energy band act as separate entities. Each process is split up into a manifold of separate, discrete processes, one for each energy band. For every energy level, energy-dependent reaction probabilities $c_\mu(E)$ must be calculated.

Three different types of reactions can be distinguished for these reaction mechanisms: (1) the initial reaction leading to the formation of energized adducts; (2) unimolecular reactions of those adducts, resulting in either isomerization or fragmentation, including redissociation into the reactants; and (3) vibrational energy transitions due to collisional energy exchange with the bath gas.

For unimolecular reactions, $c_\mu(E)$ is equal to the unimolecular rate constant $k_\mu(E)$. The number of distinct reaction combinations h_μ of an entity $S(E)$ is simply the number of reactant molecules $N_{S(E)}$ for that reaction present in the considered volume, and with the considered energy content E . The rate constants $k_\mu(E)$ can be calculated with a number of kinetic theories. In the URESAM computer program, the unimolecular rate constants $k_\mu(E)$ are calculated using RRKM theory,¹⁻⁴ because of its generally good agreement with experimental data:

$$k_\mu(E) = \frac{\alpha G^\ddagger(E - E^\ddagger)}{h\rho(E)},$$

where α is the reaction path degeneracy, $\rho(E)$ is the vibrational energy density of states of the reactant, and $G^\ddagger(E - E^\ddagger)$ is the sum of states in the energy range up to $E - E^\ddagger$ for a transition state with a barrier height $E^\ddagger$; both $\rho(E)$ and $G^\ddagger(E - E^\ddagger)$ are calculated using the Beyer–Swinehart–Stein–Rabinovitch direct-count algorithm.¹¹

Vibrational energy transitions of an activated species $S(E)$ by collisions with the bath gas M are incorporated in a similar fashion: the “reaction” probability c_μ is equal to the product of the collision number Z_{LJ} and the considered volume V . Since collisions are bimolecular reactions, h_μ is equal to $N_{S(E)} N_M$, with $N_{S(E)}$ and N_M being the number of $S(E)$ species and bath gas molecules, respectively. Energy loss or gain during this collision can be simulated by randomly selecting a transferred energy value for that collision. The probability density function P_{transfer} for this quantity is governed by the adopted collisional energy transfer model. In our calculations, we use the biexponential model as proposed by Troe.¹² According to this model, small amounts of energy are preferentially transferred, with an exponentially decreasing probability for larger amounts of transferred energy. Note that the probabilities also depend on the energy content E of the colliding $S(E)$ species through the common normalization factor Q :

$$P_{\text{transfer}}(E \rightarrow E') = \frac{1}{Q} \exp\left(\frac{E - E'}{\langle \Delta E \rangle_{\text{up}}}\right), \quad \text{for } E' > E,$$

$$P_{\text{transfer}}(E \rightarrow E') = \frac{1}{Q} \exp\left(\frac{(E' - E)}{\langle \Delta E \rangle_{\text{down}}}\right), \quad \text{for } E' < E.$$

The parameters $\langle \Delta E \rangle_{\text{up}}$ and $\langle \Delta E \rangle_{\text{down}}$ can be derived from the average transferred energy per collision $\langle \Delta E \rangle$: it is easily shown¹² that $\langle \Delta E \rangle = \langle \Delta E \rangle_{\text{up}} - \langle \Delta E \rangle_{\text{down}}$; the relation between $\langle \Delta E \rangle_{\text{up}}$ and $\langle \Delta E \rangle_{\text{down}}$ is dictated by detailed balancing:

$$\langle \Delta E \rangle_{\text{up}} = \frac{\langle \Delta E \rangle_{\text{down}} F_E kT}{F_E kT + \langle \Delta E \rangle_{\text{down}}},$$

where F_E is the energy dependence factor of the density of states:

$$F_E(E) = \frac{\int_0^\infty dE' Z(E') \rho(E') \exp(-E'/kT)}{kT Z(E) \rho(E) \exp(-E/kT)},$$

with $Z(E)$ the (Lennard–Jones) collision frequency. F_E is weakly energy-dependent but can be approximated¹² by $F_E \cong F_E(E^\ddagger)$, with $E^\ddagger$ the energy at the top of the energy barrier for reaction. The URESAM program allows for a separate probability function $P_{\text{transfer}}(E \rightarrow E')$ for each of the various energized species involved in the system of unimolecular reactions. The continuous probability functions P_{transfer} above are rewritten for use in a finite, grained energy space.

Generation of chemically activated species $S(E)$ occurs through an initial reaction $A+B$ and/or by unimolecular isomerizations of other activated species present. The stochastic reaction probability c_μ for an initial reaction $A+B$ is equal to the product of the second-order rate constant k_{ini} and

the volume V , multiplied by 2! if $A = B$; simple combinatorial arguments show that the number of distinct reactant pairs $h_\mu = N_A N_B$ if $A \neq B$, and $h_\mu = (1/2!) N_A (N_A - 1)$ if $A = B$. If the reaction of A with B can occur through multiple reaction paths forming different isomeric adduct forms—e.g., by insertion and (cyclo)addition—the initial reaction can be split into independent reactions, starting from the same molecules A and B , but each forming a different adduct i with its respective partial rate constant $k_{\text{ini},i}$. The energies of the initial reactants A and B are assumed to be Maxwell–Boltzmann distributed. Due to these thermal effects the adducts will be formed and statistically distributed over the accessible energy levels according to an energy distribution probability density function $P_{\text{form}}(E)$. This function $P_{\text{form}}(E)$ can be derived² from detailed balance considerations:

$$P_{\text{form}}(E) \delta E = \frac{G^\ddagger(E - E^\ddagger) \exp\left(-\frac{E - E^\ddagger}{kT}\right) \delta E}{\int_{E^\ddagger}^\infty G^\ddagger(E' - E^\ddagger) \exp\left(-\frac{E' - E^\ddagger}{kT}\right) dE'}$$

for $E \geq E^\ddagger$; 0 otherwise,

where $E^\ddagger$ is the barrier height for the redissociation of S^* to $A + B$; $G^\ddagger(E - E^\ddagger)$ is the sum of states for the transition state in the energy range up to $E - E^\ddagger$. Allowance for tunnelling through the barrier can be made by an Eckart-type function.¹³

E. Time evolution of the population during the simulation

The product distribution of any chemically activated system of unimolecular reactions—however complex—can then be calculated by simulating a large number of initial reaction occurrences and the subsequent isomerization, (re)dissociation, and energy transfer processes. During this calculation, four different time zones can be distinguished. During the first period, as a result of the formation of energized adducts through the initial reaction, the concentrations of these energized adducts and their isomers build up until they reach (quasi)steady state at the end of the first time zone. During the second period, formation of energized adducts in each energy band is in balance with unimolecular dissociation, isomerization, and collisional energy transitions. The stabilized molecules have a relatively long lifetime; thermal reexcitation of stabilized molecules has therefore only a negligible influence on the product distribution. However, as concentrations of the stabilized molecules increase—during the third time period—reactions of thermally reactivated molecules become increasingly important, until they also reach steady state in the fourth time zone.

Recombination of dissociation products is ignored, since it is not relevant to the product distribution of the original reaction. For the same reason, calculations of product distributions during the third and fourth time period are of limited use. The thermal rate constants for the unimolecular reactions are easily obtained in a separate calculation. This can

be done for the high pressure limit by averaging the energy-dependent rate constant $k_{\mu}(E)$, giving each energy level a Maxwell–Boltzmann fractional weight; at lower pressures one should use more sophisticated methods^{3,4} that take into account the depletion of the high-energy tail of the Maxwell–Boltzmann distribution. These rate constants can then be used in separate kinetic modeling studies to describe the thermal reactions of the stabilized fraction obtained from the calculations for the second time period.

The second time period basically gives us the product distribution of the “overall” reaction and the steady-state distributions of the intermediate molecules at the considered temperature and pressure. Obviously, meaningful results can only be obtained when, for each of the activated species, steady-state distributions have been reached. Only then will there be convergence for the relative importance of each reaction channel. Since thermal re-excitation of stabilized molecules is negligible, it is sufficient to achieve a steady-state distribution for those energy levels above and slightly below the lowest accessible transition state for that species. Molecules with a lower energy content can then be collected in a “sink” without significantly altering the calculated product distribution. This also forces the calculation to remain in the second time period, eliminating unnecessary simulation of collisions of stabilized species in the process. The steady-state energy distributions for each of the activated species provide sufficient information to calculate the relative importance of each exit channel and the fractions of the various stabilized adducts. Average rate constants for the various unimolecular reactions can also be obtained in such steady-state conditions; in calculating the average rate constant each energy level should be given a weight equal to its fraction in the steady-state population.

F. Remarks

The ESM described above requires the simulation of a large number of reactions; efficient use of computer time is therefore an important issue. The exact number of reaction simulations needed depends on the reaction mechanism and conditions, as well as the required precision of the output. For competing reactions, the speed of convergence of the ratio of individual reaction occurrences is directly related to the difference in probability (rate constants) of the competing reactions: one needs to simulate a very large total number of reactions to obtain a statistically significant number for a competing reaction with a very low probability. To address this problem under steady-state conditions, the URESAM program also calculates the time-averaged populations for each of the adducts and isomers, and the energy-distribution averaged rate constants k_{uni} for this population, using the RRKM-rate constants $k(E)$. The ratio of reaction occurrences for two unimolecular reactions under steady-state conditions should be equal to the ratio of these rate constants. The relative importance of a highly improbable reaction can therefore be calculated using the steady-state population; it is not even necessary for this reaction to actually occur in the simulation.

Another important consideration is the choice of the uniformly distributed random number generator. An unfortunate choice can strongly bias the results of the simulation. Important aspects here are the number of significant bits, uniformness of the distribution, and correlation between successive random numbers. At present, we use a combination of two multiplicative linear congruential generators,¹⁴ which induces no detectable errors in our implementation of the simulation algorithm.

III. DIRECT CALCULATION OF THE PRODUCT DISTRIBUTION (DCPD)

A. Introduction

Assume it were possible, for a given unimolecular reaction mechanism and reaction conditions, to calculate the probability $P_{X(E) \rightarrow Y}$ that an *initially* formed (adduct) species $X(E)$ would ultimately end up as a reaction product Y , taking into account all possible pathways leading to that end product Y . This information would allow the direct calculation of the product distribution (DCPD) of a chemically—or otherwise—activated chemical system of unimolecular reactions in steady-state conditions. Indeed, the overall product fraction of Y is the weighted sum of the probabilities for all initially formed species $X(E)$:

$$\text{Fraction } Y = \sum_{X(E)} (P_{\text{form}}(X(E)) \times P_{X(E) \rightarrow Y}),$$

with P_{form} the probability for the initial formation of $X(E)$ in the activation process.

B. Calculation of $P_{X(E) \rightarrow Y}$

Surprisingly, calculation of $P_{X(E) \rightarrow Y}$ can be done efficiently using simple matrix algebra, even though there are an infinite number of pathways—each with its own probability—leading from $X(E)$ to Y . The transition from $X(E)$ to Y is essentially a random walk through the energy–species space generated by all possible unimolecular reactions and collisional energy transfer reactions. This random walk can be described using a discrete time Markov chain: the relative reaction probabilities are independent of time, and we are only interested in *which* end product is formed, not *when* it is formed. This Markov chain is described by the one-step transition matrix⁹ $\mathbf{P}$, a matrix containing the probabilities $P_{i \rightarrow j}$ that a molecule in state “ i ” will next react to a state “ j .” The probability $P_{i \rightarrow j}$ of a reaction, be it a unimolecular reaction or an energy transfer reaction, can be calculated as

$$\mathbf{P}_{ij} = P_{i \rightarrow j} = \frac{c_{i \rightarrow j/k}}{\sum_{k=1}^n c_{i \rightarrow k}},$$

with $c_{i \rightarrow j/k}$ the reaction probability constant for the reaction leading from state i to state j/k (see Sec. II B), and n possible reactions originating in state i . Since end products do not react any further, we have for i the state number for the end product: $\mathbf{P}_{ii} = 1$ and $\mathbf{P}_{ij} = 0$ ($i \neq j$). These so-called absorbing states collect all the molecules reaching an end-

product state. End products are defined as those products formed by dissociation reactions, and adducts sufficiently stabilized to ensure that the probability for thermal re-excitation is negligible. This is comparable to the sink described in Sec. II E. The probability of reaching state j , starting from state i after (at most) an infinite number of reactions—to include all possible intermediate transitions—can be calculated as $(\mathbf{P}^\infty)_{ij}$ (Chapman–Kolmogorov equation⁹). Rather than calculating the infinite power of matrix $\mathbf{P}$, we decompose $\mathbf{P}$ as $\mathbf{P} = \mathbf{D}\mathbf{A}\mathbf{D}^{-1}$, with $\mathbf{D}$ a matrix containing the eigenvectors, and $\mathbf{A}$ a diagonal matrix containing the eigenvalues. Then $\mathbf{P}^\infty = \mathbf{D}(\mathbf{A}^\infty)\mathbf{D}^{-1}$, which is straightforward to calculate since eigenvalues λ of a transition matrix $\mathbf{P}$ are always $|\lambda| \leq 1$. There will be as many eigenvalues equal to 1 as there are absorbing states; both the eigenvalues and the eigenvectors can contain complex numbers.

C. Discussion

We now have $(\mathbf{P}^\infty)_{ij} = P_{X(E) \rightarrow Y}$, with i the state number of $X(E)$, and j the state number of end product Y ; this allows us to calculate the product distribution using the formula given above. Calculation of $P_{X(E) \rightarrow Y}$ is also possible using the simulation algorithm, but this requires a separate calculation for each possible $X(E)$ and is therefore unwieldy and time consuming.

The main virtue of the DCPD method is that it gives detailed information about the relationship between reactant and end product, using only simple matrix algebra. Implementation of this method is almost trivial using proven matrix computer libraries. Furthermore, $P_{X(E) \rightarrow Y}$ is independent of the initial reaction, so multiple initial reactions can be examined in one calculation. This method is only applicable in steady-state conditions or for the cumulative product distribution of a completed reaction.

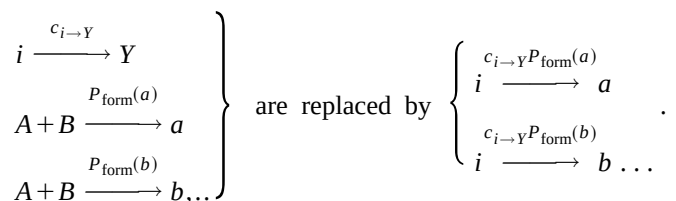
IV. CALCULATION OF THE STEADY-STATE POPULATIONS FOR THE INTERMEDIATES (CSSPI)

A. Introduction

While the previous DCPD method provides information about the product distribution, it does not give any information about the (relative) concentration of the intermediates. These concentrations fluctuate randomly around their steady-state average. Each individual molecule performs a random walk through the energy–species space, independent of the other molecules. The number of molecules active at one time depends both on the rate of the initial reaction and the rate of the removal reactions. If we calculate the probability of finding a molecule in a particular state i —the steady-state population distribution—and multiply this probability by the average number of activated intermediates (per volume), we find the average steady-state concentrations for the intermediates.

B. Calculation of the steady-state distribution

The probability of finding a molecule in a particular state also depends on the lifetime of the state. Therefore, we must rigorously include the absolute reaction probability per unit time of all processes, rather than only the relative probabilities of the reactions. The random walk can be described using a continuous time Markov chain. To calculate its steady-state distribution, we first construct the Markov chain's infinitesimal generator⁹ $\mathbf{A}$. This matrix contains the time-dependent reaction probability constants $\mathbf{A}_{ij} = c_{i \rightarrow j}$ with $c_{i \rightarrow j}$ the reaction probability constant for the reaction leading from state i to state j ; $\mathbf{A}_{ii} = -\sum_j c_{i \rightarrow j}$, the total reaction probability for destruction of a species in state i . End-product states are not included here; to ensure steady-state conditions, all end products formed are immediately “transformed” into initial adducts and reintroduced into the system. This is implemented by replacing all reactions leading to end product Y by a set of reactions also describing the initial formation process [$P_{\text{form}}(x)$ is the probability density function for formation of x]:



The resulting closed system is independent of the rate constant of the initial reaction; we can therefore initially only derive the relative concentrations. If one uses a sink, stabilized molecules will be considered an end product; otherwise, the stabilized molecules are allowed to achieve a thermal energy distribution. The number and distribution of molecules then also depends on the rate of thermal re-excitation.

The steady-state distribution is then calculated as the solution of the following set of equations:⁹

$$\mathbf{X}\mathbf{A} = \mathbf{0},$$

$$\sum \mathbf{X}_i = 1,$$

$$\mathbf{X}_i \geq 0,$$

where rowvector $\mathbf{X}$ contains the steady-state distribution; $\mathbf{X}_i$ is the steady-state fraction for state i ; and rowvector $\mathbf{X}$ is essentially the normalized eigenvector of eigenvalue $\lambda = 0$.

C. Calculation of the number of molecules per volume

The absolute number of active intermediates (per unit volume) at one time depends on both the rate of formation (through the initial reaction) and the rate of destruction (through formation of end products). The initial reaction can be described as a Poisson process with intensity $c_{\text{ini}} h_{\text{ini}}$ (see Sec. II B). The destruction reactions themselves are also Poisson processes; their intensity, however, changes due to the stochastic fluctuations of the distribution and the total

number of molecules in the system. Still, the intensity of a unimolecular destruction process is linearly dependent on the number of molecules (see Sec. II A) in its originating state i , so, given a sufficient number of molecules and/or long enough reaction times, the number of destruction reactions is indistinguishable from a true Poisson process with intensity $c_{i \rightarrow Y} N_i = c_{i \rightarrow Y} \mathbf{X}_i N_{\text{tot}}$ (N_i is the number of molecules in state i , N_{tot} is the total number of activated molecules in the system). Combining all destruction processes gives rise to a Poisson process with intensity $\sum_i \sum_Y (c_{i \rightarrow Y} \mathbf{X}_i N_{\text{tot}})$. Steady state implies that the intensities of the Poisson processes for formation and destruction are equal; this gives us an expression to calculate N_{tot} :

$$c_{\text{ini}} h_{\text{ini}} = \sum_i \sum_Y (c_{i \rightarrow Y} \mathbf{X}_i) \times N_{\text{tot}}.$$

D. Discussion

Once we have the steady-state distribution $\mathbf{X}$ with $\mathbf{X}_i$ the steady-state fraction for each state i , we can use this information to calculate the energy-distribution averaged rate constants for each reaction $S \rightarrow T$ and the global product distribution:

$$k_{S \rightarrow T} = \frac{\sum_E (c_{S(E) \rightarrow T(E)} \mathbf{X}_{S(E)})}{\sum_E \mathbf{X}_{S(E)}},$$

$$\text{Fraction } Y = \frac{\sum_i c_{i \rightarrow Y} \mathbf{X}_i}{\sum_i \sum_Y (c_{i \rightarrow Y} \mathbf{X}_i)}.$$

Multiplication of $\mathbf{X}_i$ with N_{tot} yields the absolute concentrations (number of molecules N_i per unit volume) of the intermediates. Again, implementation of this algorithm is trivial, requiring only the calculation of one eigenvector for a known eigenvalue (0 is always an eigenvalue of the infinitesimal generator $\mathbf{A}$ for a finite conservative irreducible continuous Markov chain). For the calculation of the overall product distribution, this CSSPI method is even faster—calculation of only one eigenvector—than the DCPD method, but does not provide the same detailed information about the origin of this product distribution. Calculating this would require a different calculation for each originating state, at the expense of the speed advantage of the CSSPI method.

If one rewrites the set of equations from Sec. IV B to include the expression for N_{tot} , and replaces all stochastic parameters c and h by their deterministic counterparts k and $[X]$, one ultimately recovers the same algebraic set of equations one would obtain by imposing strict steady state on deterministic rate equations:

$$\frac{d[X_1]}{dt} = k_{\text{ini}1}[A][B] + \sum k_{j \rightarrow 1}[X_j] - \sum k_{1 \rightarrow j}[X_1] = 0,$$

$$\frac{d[X_2]}{dt} = k_{\text{ini}2}[A][B] + \sum k_{j \rightarrow 2}[X_j] - \sum k_{2 \rightarrow j}[X_2] = 0,$$

and so on. It is reassuring to see both descriptions—stochastic and deterministic—yield the same average steady-state concentrations. This does not mean, however, that both are in general equivalent. The stochastic description inher-

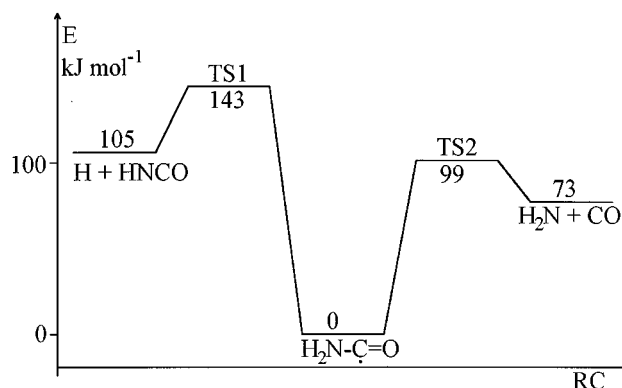


FIG. 1. Schematic potential energy profile of the H+HNCO reaction.

ently describes chemical reactions on a physically sounder basis, using randomness and statistic fluctuations as starting points, rather than only trying to describe the average changes in a chemical mixture. Still, apart from these conceptual and some minor mathematical differences—which can be neglected for realistic systems¹⁵—the calculations using continuous Markov chains are comparable to methods derived from the matrix formulation of sets of linear differential equations.^{3,4}

V. SAMPLE APPLICATION OF THE URESAM COMPUTER PROGRAM TO A SIMPLE SYSTEM, AND COMPARISON OF THE THREE PRESENTED METHODS

We will next apply all three methods to the H+HNCO reaction. Detailed calculations on this system have been published earlier,¹⁶ using an energy band size of 0.5 kJ mol^{−1} and a maximum energy of 600 kJ mol^{−1} above the ground state of H₂NCO. The calculations presented here are primarily illustrations of the possibilities of the URESAM program, using an energy band size of 1 kJ mol^{−1} and a maximum energy of 450 kJ mol^{−1}. The chemically activated H+HNCO reaction mechanism is shown in Fig. 1; the initially formed H₂NCO⁺ can (re)dissociate to form either H₂N+CO or H+HNCO, or be stabilized as H₂NCO. The rate constants of the unimolecular reactions are calculated using the RRKM theory and energy transfer reactions are modeled using the biexponential model of Troe. All relevant data have been published earlier.¹⁶ For clarity, no tunneling is included. The cut-off level for the sink is 40 kJ mol^{−1} below the lowest transition state TS2. The calculations discussed here are performed for a temperature of 1500 K and a pressure of 100 atm. At this temperature the rate constant¹⁶ of the initial H+HNCO reaction is approximately 2.2×10^{-12} molecules cm^{−3} s^{−1}. With initial H and HNCO concentrations of 10¹⁵ molecules cm^{−3}, and a considered reaction volume V of 1 cm³, the initial reaction is set to occur on average $h_{\text{ini}} \times N_H \times N_{\text{HNCO}} = 2.2 \times 10^{18}$ times per second. Depletion of the initial reactants is negligible up to a time of 10^{−6} s. More results on the temperature and pressure dependences are shown in the comparison (Table I). All timings are given for a 67 MHz IBM RS/6000 model 3BT.

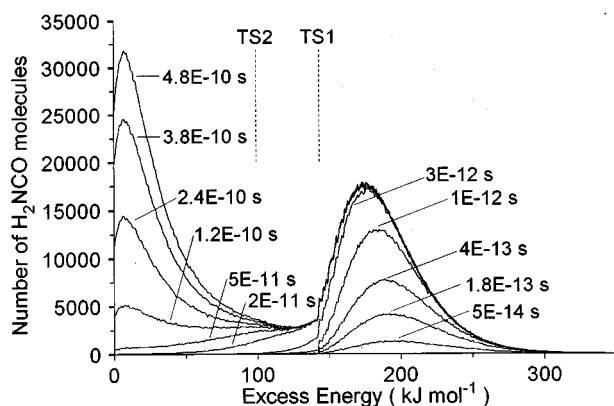
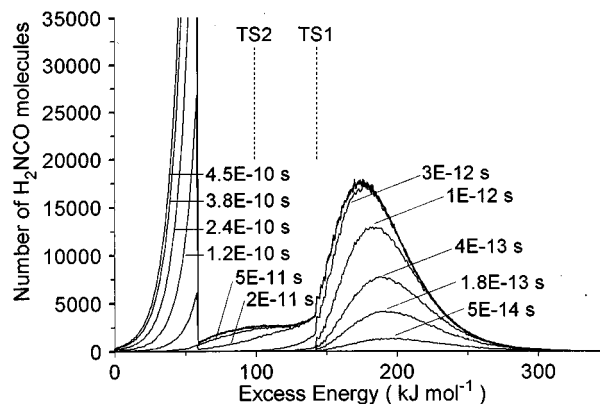
TABLE I. Temperature and pressure dependence of the product distribution (%) of the H+HNCO reaction, calculated with ESM, DCPD, and CSSPI, respectively.

Method	1 atm.			100 atm.			1000 atm.		
	H ₂ NCO	H ₂ N+CO	H+HNCO	H ₂ NCO	H ₂ N+CO	H+HNCO	H ₂ NCO	H ₂ H+CO	H+HNCO
500 K									
ESM	3.07E-4	99.2909	0.7087	5.44647	93.9442	0.609268	56.1384	43.5670	0.294566
DCPD	2.97E-4	99.2905	0.709229	5.44672	93.9442	0.609118	56.1238	43.5818	0.294313
CSSPI	2.97E-4	99.2905	0.709229	5.44672	93.9442	0.609118	56.1238	43.5818	0.294313
1000 K									
ESM	1.38E-4	96.6720	3.32782	0.67066	96.0702	3.25912	17.1270	80.0752	2.79783
DCPD	1.40E-4	96.6713	3.32857	0.66988	96.0713	3.25880	17.1386	80.0669	2.79450
CSSPI	1.40E-4	96.6713	3.32857	0.66988	96.0713	3.25880	17.1386	80.0669	2.79450
1500 K									
ESM	7.52E-5	92.5747	7.42525	0.145745	92.4658	7.38848	4.75861	88.1518	7.08954
DCPD	7.99E-5	92.5733	7.42665	0.145547	92.4666	7.38783	4.75818	88.1527	7.08910
CSSPI	7.99E-5	92.5733	7.42665	0.145547	92.4666	7.38783	4.75818	88.1527	7.08910
2000 K									
ESM	5.97E-5	88.0433	11.9566	0.051633	88.0140	11.9343	1.65032	86.5866	11.7631
DCPD	5.81E-5	88.0416	11.9583	0.051725	88.0119	11.9364	1.65061	86.5844	11.7649
CSSPI	5.81E-5	88.0416	11.9583	0.051725	88.0119	11.9364	1.65061	86.5844	11.7649
3000 K									
ESM	5.03E-5	80.4193	19.5806	0.017970	80.4089	19.5730	0.444357	80.0536	19.5019
DCPD	5.14E-5	80.4145	19.5854	0.017931	80.4066	19.5755	0.443919	80.0542	19.5018
CSSPI	5.14E-5	80.4145	19.5854	0.017931	80.4066	19.5755	0.443919	80.0542	19.5018

A. The ESM

Due to the large energy difference between the entrance transition state TS1 and the lowest exit channel transition state TS2, stabilization of H₂NCO is an unlikely event, even at moderately high pressures. Calculation for a pressure of 100 atm and 1500 K (Fig. 2) shows that steady-state conditions for chemically active H₂NCO molecules—with energies above (or slightly below) the lowest transition state—are reached early on in the reaction. Accumulation of stabilized H₂NCO-molecules is relatively slow. Of the 5.3×10^8 initially formed H₂NCO adducts after a reaction time of 2.4×10^{-10} s, 7.4% redissociated to H+HNCO and 92.3% dissociated to H₂N+CO. Of the remaining H₂NCO molecules, 0.1% can be considered stabilized—those molecules with energy content at least 40 kJ mol⁻¹ below the lowest transition state—while the others (0.2%) are still ac-

tive. As can be seen, the stabilized molecules slowly adopt a Maxwell–Boltzmann distribution. Thermal reexcitation is a negligible process at short reaction times, as illustrated in Fig. 3: we see that using a sink does not alter the population up to a time of 5×10^{-11} s for the chemically relevant energies, and therefore does not alter the product distribution significantly during the reaction times under consideration. Due to the fairly shallow energy minimum, only 99 kJ mol⁻¹, the second time period is relatively short: further calculations, using no sink, show that the thermal reactions of H₂NCO gradually become more important, and the population reaches steady state for *all* energies at times as short as 10^{-7} s (not shown). This indicates that the thermal reactions of the molecules collected in the sink should be taken into account for realistic reaction times. At lower pressures—e.g., 1 atmosphere for practical applications of

FIG. 2. Time-dependent population of H₂NCO⁺ formed in the H+HNCO reaction at 100 atm and 1500 K.FIG. 3. Same as Fig. 2, but with a sink “freezing” all H₂NCO molecules with energies < 60 kJ mol⁻¹.

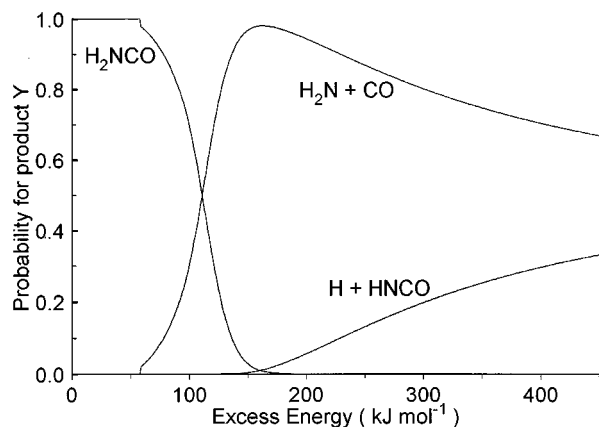


FIG. 4. Probability $P_{X(E) \rightarrow \text{product } Y}$ for formation of each possible product Y as a function of the energy E of the initially formed $\text{H}_2\text{NCO}^\ddagger$ adduct, for a temperature of 1500 K and a pressure of 1 atm.

RAPRENO_x¹⁷—and likewise at lower temperatures, thermal reactivation will be less important. Note that the structure in Figs. 2 and 3 is not noise, it results from the discontinuous behavior of the unimolecular $k(E)$ and of $P_{\text{form}}(E)$.

The ESM can be used to derive any property; for most applications we are only interested in the product distribution, the time needed to achieve steady state, and the steady-state concentrations of the intermediates. For the mechanism presented here, the current version of URESAM simulates over 1.1 million events (total) per CPU minute.

B. The DCPD method

We then calculate the product distribution of the $\text{H} + \text{HNCO}$ directly using the DCPD method; we assume 100 atm and 1500 K. Figure 4 gives the probabilities of finding one of three different end products $\text{H} + \text{HNCO}$, $\text{NH}_2 + \text{CO}$ and $\text{H}_2\text{NCO}_{\text{stab}}$, given the initial energy of $\text{H}_2\text{NCO}^\ddagger$. Multiplication of this probability with the formation energy probability density function $P_{\text{form}}(E)$ (see Sec. II D) in Fig. 5 gives us information about the origin of the products: most of the stabilized molecules were initially formed with low energy content. Formation of $\text{NH}_2 + \text{CO}$ closely follows the formation energy probability density function $P_{\text{form}}(E)$; formation of $\text{H} + \text{HNCO}$ is shifted to somewhat higher initial energies. This information would be more interesting for complex systems. Calculation of the overall product distribution, as described in Sec. III, yields 7.4% redissociation to $\text{H} + \text{HNCO}$, 0.1% stabilization, and 92.5% dissociation to $\text{NH}_2 + \text{CO}$. The total calculation required 19 s of CPU time.

We can also use the already calculated matrix $\mathbf{P}^\infty$ to derive the product distribution of the $\text{H}_2\text{N} + \text{CO}$ reaction under the same reaction conditions (90.3% redissociation to $\text{H}_2\text{N} + \text{CO}$, 7.4% stabilization, and 2.3% dissociation to $\text{H} + \text{HNCO}$) with negligible overhead.

C. The CSSPI method

The steady-state energy distribution of the intermediates at a pressure of 100 atm and a temperature of 1500 K is

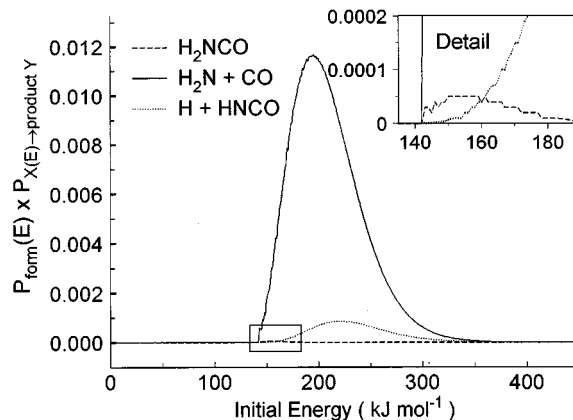


FIG. 5. The contribution of an energy band (here 1 kJ mol⁻¹ wide) with energy E , to the total product distribution of the $\text{H} + \text{HNCO}$ reaction at 1500 K and 100 atm is calculated by multiplying the probability $P_{X(E) \rightarrow \text{product } Y}$ with the probability of formation of a H_2NCO adduct within that energy band, $P_{\text{form}}(E)$. Summing the contributions from all energy bands gives the overall yield for the considered product.

shown in Fig. 6. The average number of active molecules in the system is 1.34×10^6 . Also note the perfect match between the steady-state population shown in Figs. 6 and 3. Even at a pressure of 100 atm, stabilization is still a fairly improbable event: most of the molecules never reach the energy region below the entrance transition state TS1. As a result, the distribution above TS1 closely resembles the formation energy density function $P_{\text{form}}(E)$. Molecules with energies between TS1 and TS2 are formed mainly through down transitions of molecules with energy higher than TS1, hence the higher concentration of molecules with energy closer to TS1. The slight increase in concentration at energies close to TS2 is due to the longer lifetime of these molecules. At energies below TS2, most molecules disappear in the sink, resulting in a decreasing concentration as energy decreases. From this distribution we calculate the product distribution, as described in Sec. IV D. We find 0.1% stabi-

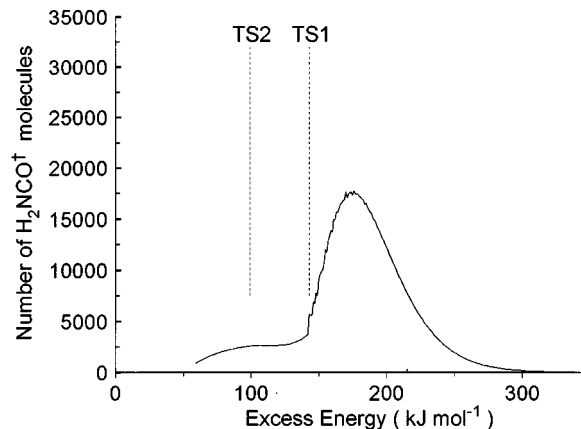


FIG. 6. Steady-state population of the activated $\text{H}_2\text{NCO}^\ddagger$ formed in the $\text{H} + \text{HNCO}$ reaction at 1500 K and 100 atm. The sink removes all H_2NCO molecules with less than 60 kJ mol⁻¹ of energy.

lization, 7.4% dissociation to $\text{H}_2\text{N} + \text{CO}$, and 92.5% redissociation to $\text{H} + \text{HNCO}$. The entire calculation required 1 s of CPU time.

D. Comparison

The results of all three methods are identical for all practical purposes. While the DCPD and CSSPI methods yield essentially complementary information—distribution of the products versus distribution of the still active molecules—the product distributions obtained differ only in the least significant digits of the machine precision, many orders of magnitude below the errors induced by the approximations of the rate theories and accuracy of the input data (energies, vibration frequencies, etc.). Also, they match the essentially exact solution of the master equation by the ESM, even though this algorithm is fundamentally different. Again, the difference in the results is much below the errors induced by theory and input data.

Table I shows the pressure and temperature dependence of the steady-state product distribution of the $\text{H} + \text{HNCO}$ reaction, calculated with the three methods. At all reaction conditions, the results compare very favorably. The ESM calculations were not allowed to achieve full convergence; this results in a slight deviation of the least significant digits as compared to the DCPD/CSSPI methods.

VI. CONCLUSION

We have presented three methods to calculate various characteristics of chemically activated unimolecular reactions, based on the stochastic description of chemical reactions. The first of these methods, the ESM, allows the full time-dependent solution of the master equation. The other two methods, DCPD and CSSPI are applicable to reactions in steady-state conditions, and yield information about the product distribution and steady-state population at greatly reduced computational cost. All methods—while based on different algorithms—are in excellent agreement, indicating that only a negligible error is induced by the calculation method itself.

As an example, we applied these methods to the $\text{H} + \text{HNCO}$ reaction—relevant for the RAPRENO_x mechanism for NO_x removal from flue gases¹⁷—at various temperatures and pressures, revealing the $\text{H} + \text{HNCO}$ reaction as an efficient H_2N source at moderate pressures and/or higher temperatures.

We intend to use these methods in future work on chemically activated unimolecular reactions.

ACKNOWLEDGMENTS

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